PARISHRAM 2.0

CLASS 12TH 2026

Botany - Sexual Reproduction in Flowering Plants

BIOLOGY | LECTURE-2

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TOPICS TO BE COVERED

1. MICROSPOROGENESIS
2. POLLEN GRAIN
3. MEGASPOROGENESIS
4.

MICROSPOROGENESIS

Flower → Androecium → (2n) Anther

(2n) — ♂ sex organ (2n)

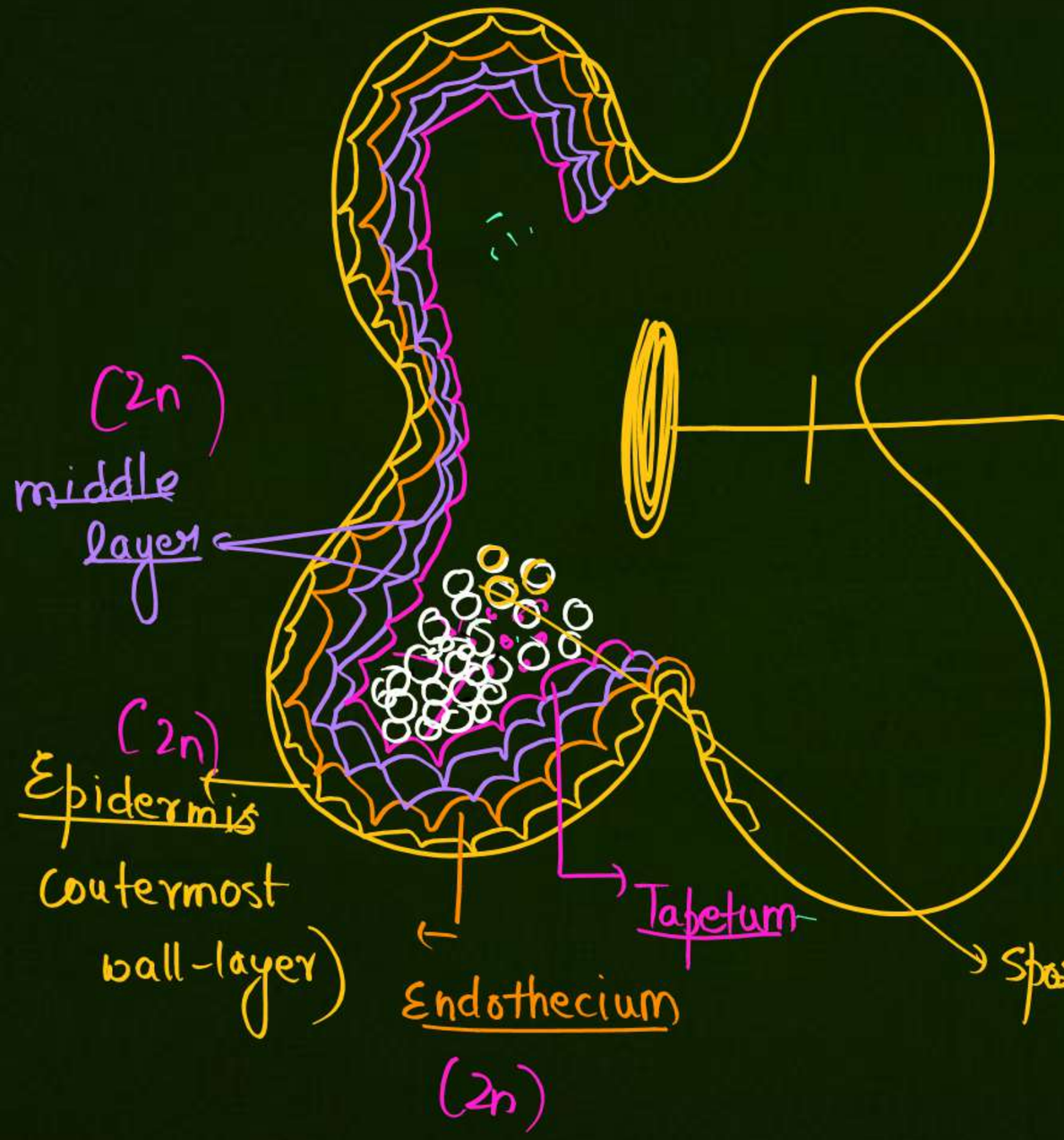
Structure of Anther

Bilobed, dithecous (mature)

{ Anther → pollen grain

Microsporogenesis : As the anther develops, the cells of the sporogenous tissue undergo meiotic divisions to form microspore tetrads. *What would be the ploidy of the cells of the tetrad?* (Haploid)

As each cell of the sporogenous tissue is capable of giving rise to a microspore tetrad. Each one is a potential pollen or microspore mother cell. The process of formation of microspores from a pollen mother cell (PMC) through meiosis is called **microsporogenesis**. The microspores, as they are formed, are arranged in a cluster of four cells – the **microspore tetrad** (Figure 1.3a). As the anthers mature and dehydrate, the microspores dissociate from each other and develop into **pollen grains** (Figure 1.3 b). Inside each microsporangium several thousands of microspores or pollen grains are formed that are released with the dehiscence of anther (Figure 1.3c).



Typically a mature anther has 4-wall layers

connective [sterile band of tissue]

Epidermis — Endothecium — Middle layer

act as protective unit

+

Helps in dehiscence of anther

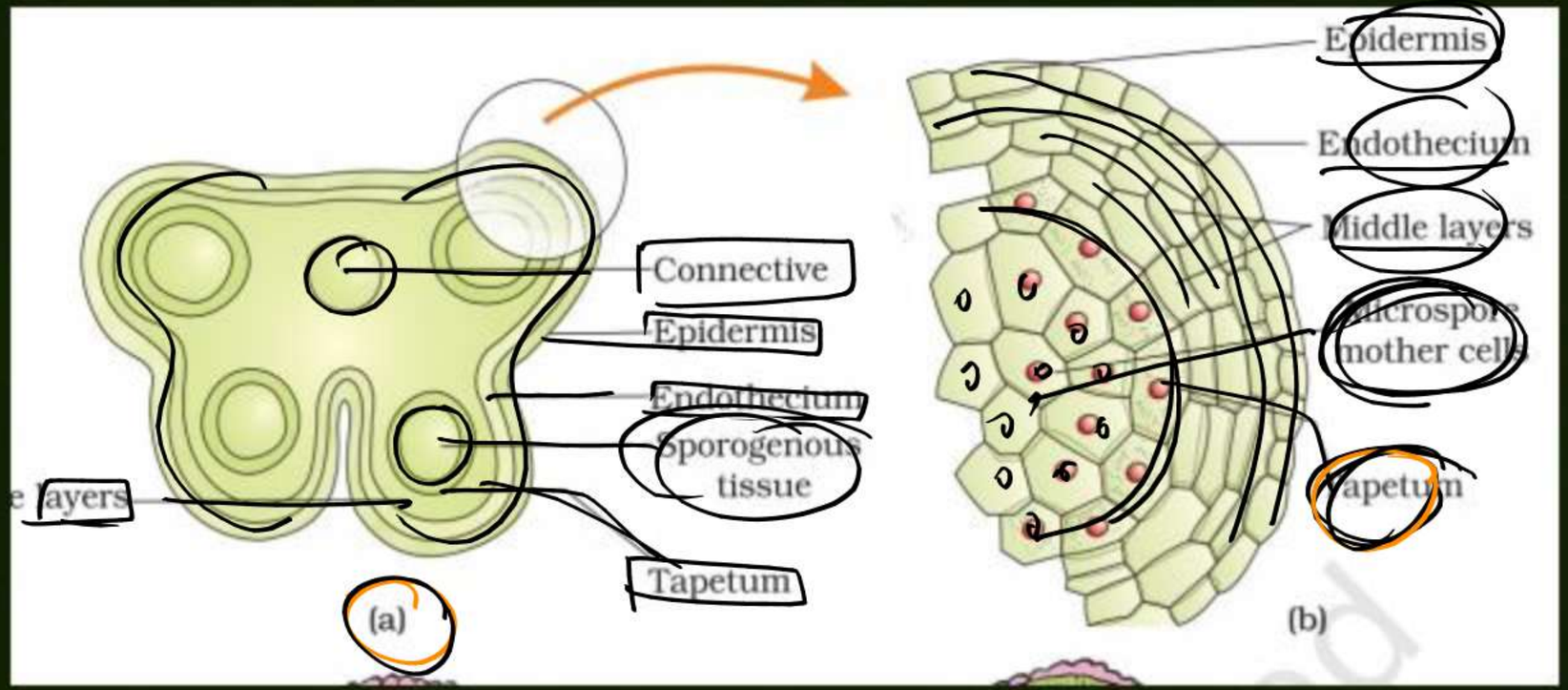
Tapetum

* Innermost layer of anther
* Cells polyploid (more than 2 sets of chromosome)
* High DNA amount
* Provide nourishment to the developing pollen grain

→ Below tapetum, is present mass of parenchymatous sporogenous tissue

+ Each sporogenous cell can behave as microspore mother cell / pollen mother cell (PMC) — 2n

→ Each PMC (2n) undergoes meiosis to produce 4 haploid (n) microspore tetrad


Connective
Epidermis
Endothecium
Sporogenous tissue
Tapetum
(a)
Epidermis
Endothecium
Middle layers
mother cells
Tapetum
(b)



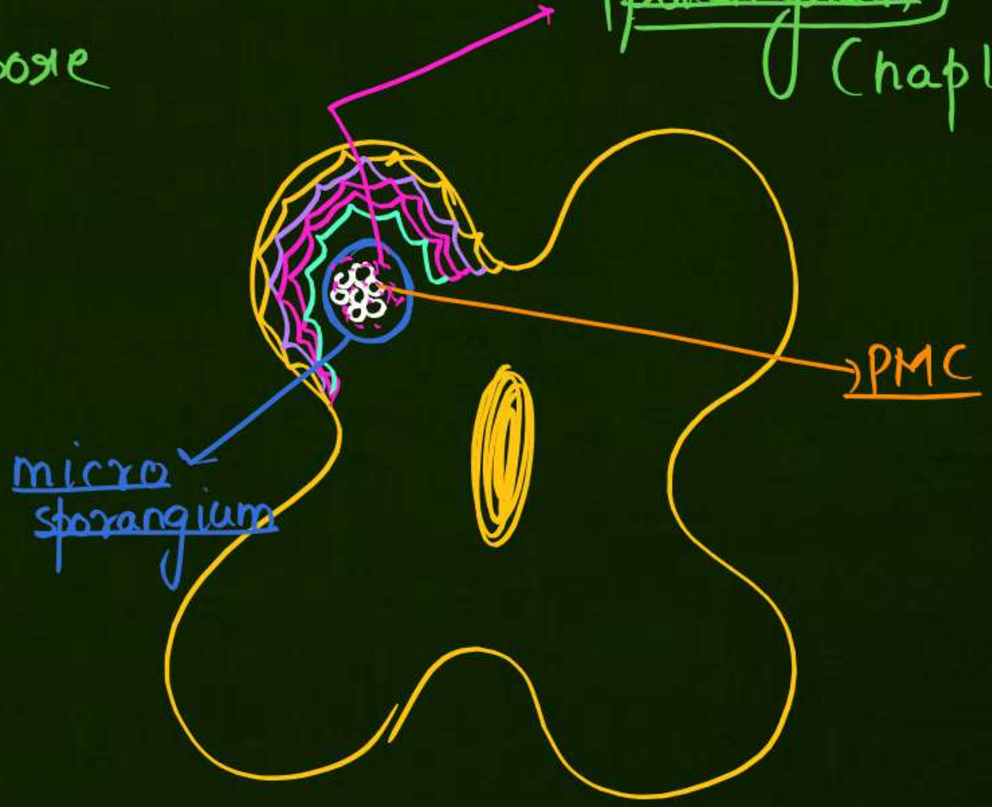
PMC/
MMC
(2n)

4, n, microspore
tetrad

Each microspore further develop into pollen grain (haploid)

Microsporogenesis

When inside anther, MMC undergoes meiosis to produce haploid microspore tetrad



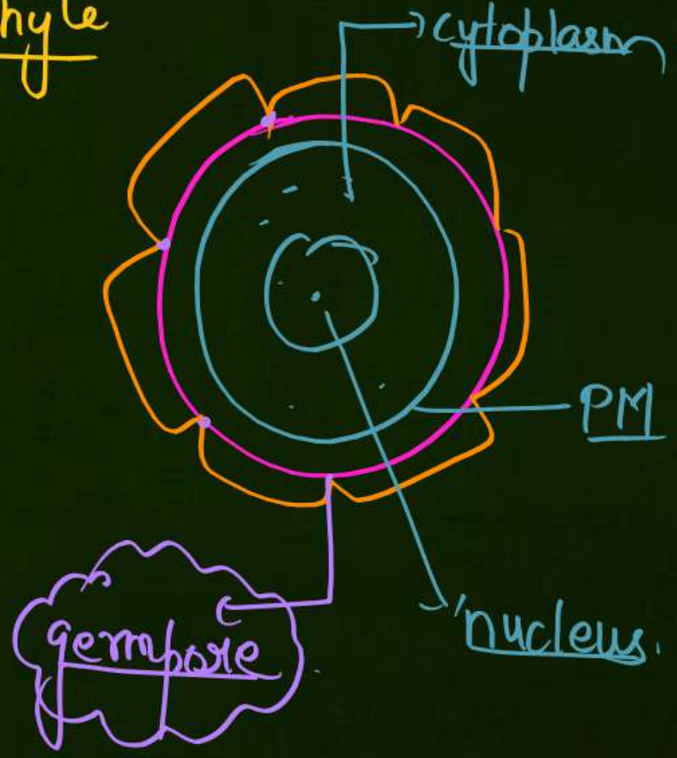
Flower → Androecium → Anther (meiosis) → pollengrain ? → ♂ gamete

Flower (2n) → Androecium / stamen (2n) → Anther (2n) — sporophyte

pollengrain (n) — male gametophyte

Structure of pollen grain :-

- → Spherical
- → 25–50 µm
- → Haploid
- → 2-wall layer = sporoderm
  - → Intine
    - → Thin, cellulose
    - → Cellulose-pectin
  - → Outer
    - → Exine
      - Thick
      - → discontinuous

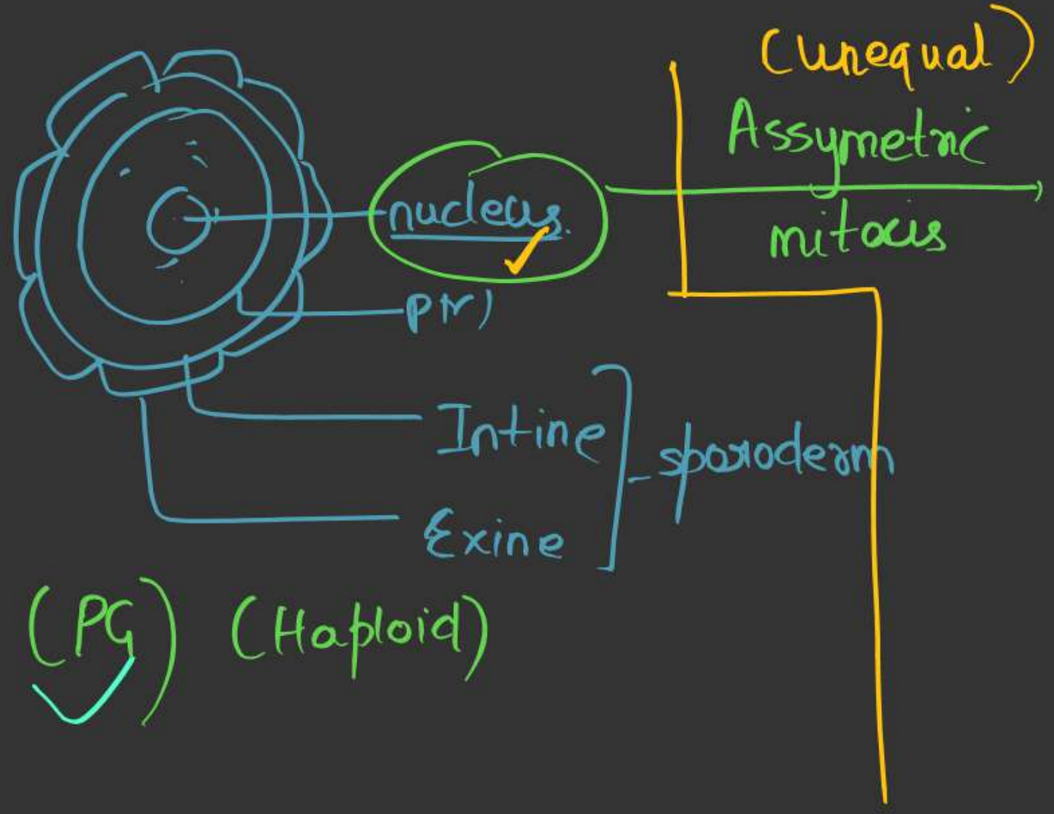


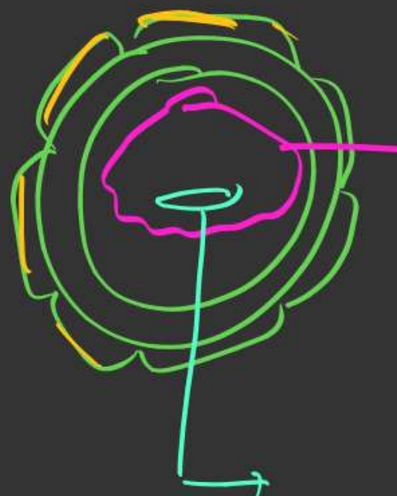
Exine → is made of hardest k/a chemical – sporopollenin.

∴ no sporopollenin ← exine absent ← germpore



Note:– Due to sporopollenin, Pollengrain cannot decomposed hence remain as fossil





Irregular-shaped, large due to more food reserve

= vegetative cell / Tube cell

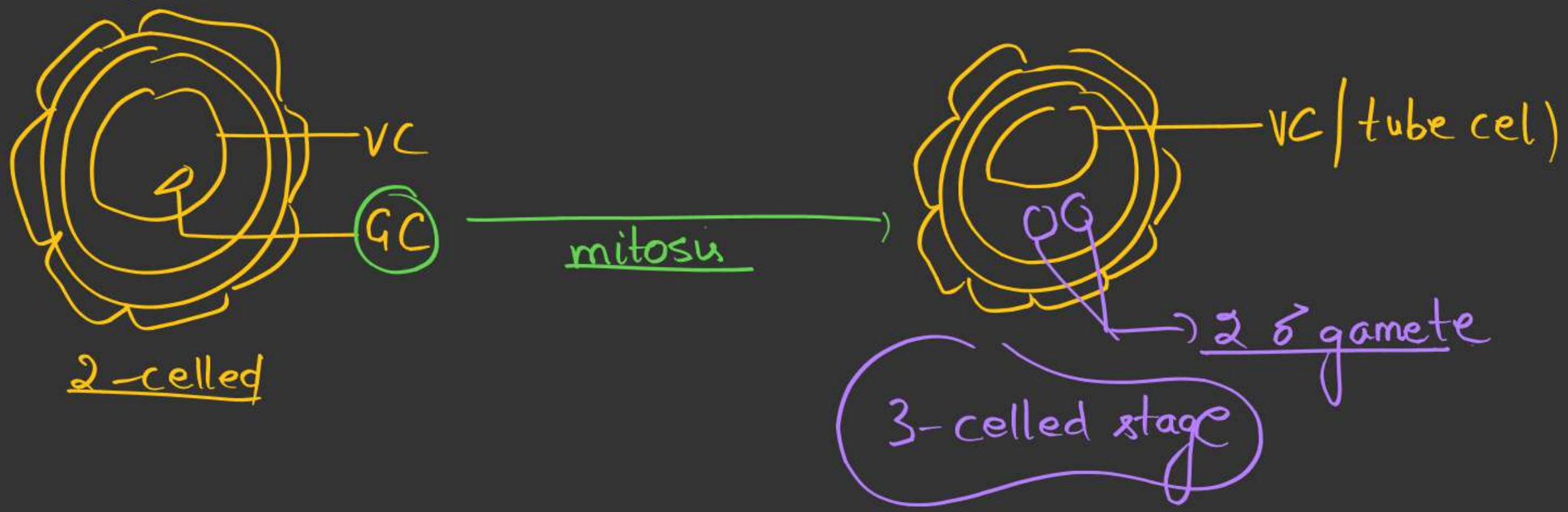
smaller spindle-shaped generative cell

- dense cytoplasm

(2-celled)

(In 60% angiospermic families at 2-celled stage of PG, anther dehisces

Remaining 40%.



2-celled

Pollen grains are generally spherical measuring about 25-50 micrometers in diameter. It has a prominent two-layered wall. The hard outer layer called the **exine** is made up of sporopollenin which is one of the most resistant organic material known. It can withstand high temperatures and strong acids and alkali. No enzyme that degrades sporopollenin is so far known. Pollen grain exine has prominent apertures called **germ pores** where sporopollenin is absent. Pollen grains are well-preserved as fossils because of the presence of sporopollenin. The exine exhibits a fascinating array of patterns and designs. *Why do you think the exine should be hard? What is the function of germ pore?* The inner wall of the pollen grain is called the **intine**. It is a thin and continuous layer made up of cellulose and pectin. The cytoplasm of pollen grain is surrounded by a plasma membrane. When the pollen grain is mature it contains two cells, the **vegetative cell** and **generative cell** (Figure 1.5b). The vegetative cell is bigger, has abundant food reserve and a large irregularly shaped nucleus. The **generative cell** is small and floats in the cytoplasm of the vegetative cell. It is spindle shaped with dense cytoplasm and a nucleus. In over 60 per cent of angiosperms, pollen grains are shed at this 2-celled stage. In the remaining species, the generative cell divides mitotically to give rise to the two male gametes before pollen grains are shed (3-celled stage).

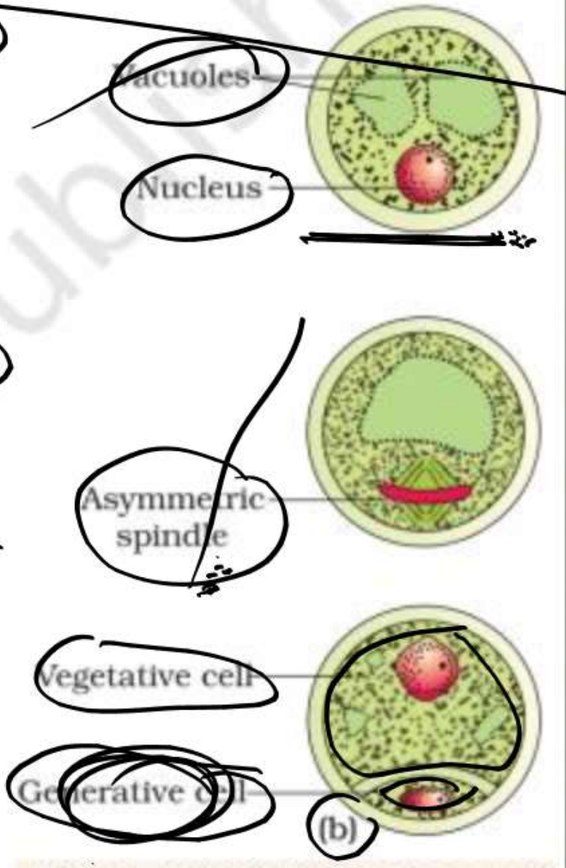


Figure 1.5 (a) Enlarged view of a pollen grain tetrad; (b) stages of a microspore maturing into a pollen grain

Pollen grains of many species cause severe allergies and bronchial afflictions in some people often leading to chronic respiratory disorders – asthma, bronchitis, etc. It may be mentioned that *Parthenium* or carrot grass that came into India as a contaminant with imported wheat, has become ubiquitous in occurrence and causes pollen allergy.

QUESTION

How many wall layers does anther has?

1. One
2. Two
3. Three
4. Four



Summary

Type Here

Q Funct of tapetum

a – Nourishment ✓

b – Protection

3 → dehiscence

d – all of the above

Q How was the class

a → Excellent

b – V. good

c – okay

d – don't know

THANK YOU